

# Sicheng Lai

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Leo-Lsc

## EDUCATION

The Chinese University of Hong Kong, Shenzhen

B.Sc. Computer Science and Engineering

Sept. 2022 – June 2026

GPA: 3.679 / 4.0

Exchange Semester in University of California San Diego

March 2025 – July 2025

## PUBLICATIONS

<sup>†</sup> Equal Contribution. <sup>\*</sup> Corresponding author.

Yadi Cao<sup>†</sup>, **Sicheng Lai**<sup>†</sup>, Jiahe Huang<sup>†</sup>, Yang Zhang<sup>†</sup>, Zach Lawrence<sup>†</sup>, Rohan Bhakta<sup>†</sup>, Izzy F. Thomas<sup>†</sup>, Mingyun Cao<sup>†</sup>, Chung-Hao Tsai, Zihao Zhou, Yidong Zhao, Hao Liu, Alessandro Marinoni, Alexey Arefiev, and Rose Yu<sup>\*</sup>

“SimulCost: A Cost-Aware Benchmark for Automating Physics Simulations with LLMs”.

[arXiv preprint](#). [Project Website](#).

**Sicheng Lai**<sup>†</sup>, Dingjie Song<sup>†</sup>, Mingxuan Wang, Shunian Chen, Lichao Sun, and Benyou Wang<sup>\*</sup>

“Both Text and Images Leaked! A Systematic Analysis of Data Contamination in Multimodal LLM”.

[ICML 2025 DIG-BUGs Oral](#). [EMNLP 2025 Findings](#).

## RESEARCH EXPERIENCE

**SimulCost: A Cost-Aware Playground**

UC San Diego | Co-First Author

April 2025 - Present

Supervisor: Prof. [Rose Yu](#)

- Developed [SimulCost](#), a comprehensive benchmarking framework evaluating LLM cost-accuracy trade-off on scientific simulation optimization across 13 physics solvers and 40+ hyperparameter tuning tasks.
- Engineered a scalable Playground API enabling on-demand simulation generation, with caching mechanisms and pandas DataFrame-based data persistence reducing redundant computations by 50%.
- Benchmarked LLMs against brute-force (e.g., grid search) and Bayesian optimization (e.g., Gaussian process) baselines across 6 real-world scenarios, conducting statistical analysis and ablation studies.

**Data Contamination in Multimodal LLMs**

CUHK-Shenzhen | Co-First Author

June 2024 - December 2024

Supervisor: Prof. [Benyou Wang](#)

- Pioneered systematic multimodal data contamination detection by designing the [MM-Detect](#) framework, incorporating two novel methods: *Option Order Sensitivity Test* and *Slot Guessing for Perturbed Caption*.
- Validated the framework’s effectiveness and sensitivity by training LLaVA-1.5-7B variants with controlled leakage at early, mid, and late training stages, and varying contamination ratios (10%, 50%, 100%).
- Developed a heuristic method to trace contamination origins, revealing unimodal contamination in base LLMs and cross-modal contamination through training data overlap analysis.

## INTERNSHIPS

**Multimodal Algorithm Engineer**

Tsinghua Shenzhen International Graduate School

December 2024 - March 2025

Shenzhen, China

- Optimized MLLM configurations through systematic experimentation with various combinations of LLMs (Qwen2.5/Vicuna) and visual encoders (CLIP/MLCD/SigLIP) based on the LLaVA-Next framework.

- Evaluated model performance using *VLMEvalKit* and *lmms-eval* to determine the optimal configuration (*Qwen2.5* + *SigLIP*), achieving top 20% ranking on the *OpenCompass* MLLM Leaderboard.
- Trained the *Qwen2.5* + *SigLIP* model on agricultural datasets curated and augmented with *data-juicer*, achieving expert-level agricultural capabilities while maintaining high general-domain performance.

## HONORS

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|---------------------------------------------------------------------|-------------|
| Dean’s List ( <i>Outstanding academic merit</i> )                   | 2022 – 2025 |
| Undergraduate Research Awards ( <i>Key research contributions</i> ) | 2024 – 2025 |

## TEACHING

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| Undergraduate Student Teaching Fellow (STA2001: Probability and Statistics I) | Spring 2024 |
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## CORE COMPETENCIES

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- **Programming & Systems:** Python, Java, C/C++ , RISC-V
- **LLM:** PyTorch, Transformers, LangChain, Accelerate, Triton, DeepSpeed, bitsandbytes, vLLM, SGLang
- **Data Science:** NumPy, pandas, SciPy, Dask, xarray, Matplotlib, Seaborn, Weights & Biases
- **Developer Tools & Deployment:** Hydra, Git, Conda, Poetry, uv, Docker, Kubernetes, SLURM, Linux
- **Backend & Web:** Flask, REST APIs, SQLite, HTML/CSS/JavaScript, TypeScript, Astro, Tailwind CSS